The Emotional State Classification using Physiological Signal Interpretation Framework

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***Abstract*—This paper proposes and evaluates an emotional state classification using a physiological signal interpretation framework. The proposed *Emo-CSI* framework consists of three components which are the following: 1) physiological signal sensing, 2) data pre-processing, and 3) emotional state classification. The *Emo-CSI* framework applies physiological signals (i.e., heart rate, breathing pattern, skin temperature, skin humidity, and skin conductivity) to classify the emotional state. The emotional state classification results in an emotional state (i.e., displeasure, neutral, pleasure, calm, medium, and excited). This research also investigates the accuracy of three classification techniques which are the following: 1) support vector machine (SVM), 2) artificial neural network (ANN), and 3) decision tree (DT). The evaluation results show that the physiological signals are related to emotional state. Using SVM as a classification in *Emo-CSI* outperforms the other classification techniques.**

***Index Terms*—emotional state, physiological signal, sensor, classification**

I. Introduction

Emotion is essential to all intentional behaviors [1]. Emotions play an important role in interpersonal communication and the decision making process. The naturalist Charles Darwin believed that emotions are adaptations that allow both humans and animals to survive and reproduce [2]. The emotion recognition is significantly applied not only to human-human interactions but also in human-computer interactions (e.g., interactive learning [3], computer games [4], and multimedia [5]).

However, the emotion recognition is not a straightforward process. The facial, voice, and gesture expressions are typically applied in emotion recognition process although these expressions can be hidden or feigned to be another emotion state [6], [7]. The ethnicity and cultures also influence on emotional recognition [8]–[10]. These challenges lead to difficulties in automatic emotional recognition research.

Instead of using facial, voice, and gesture expressions, this research paper proposes to apply physiological signals. The physiological signals are generated by autonomous nervous system (ANS). That makes humans unable to pretend or conceal their emotions [3]. The physiological signals are also less sensitive to ethnicity and cultures [11]. The physiological signal interpretation is one of the most precise way to recognize emotions [6]. There are several physiological signals those are involved with emotion such as heart rate, breathing/respiratory rate, skin conductivity, skin temperature, skin humidity, electromyogram (EMG), and electroencephalograms (EEG) [3], [7], [12]. However, these physiological sensors are expensive and may not be suitable for automatic emotion recognition.

The proposed framework, called *Emo-CSI* (the ***Emo***tional ***C***lassification using physiological ***S***ensor ***I***nterpretation), automatically detects the emotional state. The proposed *Emo-CSI* framework consists of three components which are the following: 1) physiological signal sensing, 2) data pre-processing, and 3) emotional state classification. The *Emo-CSI* framework applies physiological signals (i.e., heart rate, breathing pattern, skin temperature, skin humidity, and skin conductivity) to interpret the emotional state. The physiological sensors are carefully designed and custom-made for emotional state classification. The data pre-processing is applied to the physiological signals to yield basic statistical data. The emotional state classification takes the statistical data as inputs and results in emotional state (i.e., displeasure, neutral, pleasure, calm, medium, and excited). This research contributes a framework for the emotional state classification by using the custom-made physiological sensors which are inexpensive and instantly applied to a human body. This research also investigates the accuracy of three classification techniques which are the following: 1) support vector machine (SVM), 2) artificial neural network (ANN), and 3) decision tree (DT).

The paper is organized in the following manner: Section II describes the emotional detection principles used in the proposed framework. Section III introduces the proposed *Emo-CSI* framework. Section IV evaluates physiological sensor signals of the *Emo-CSI* framework and investigates the accuracy of the classification techniques. Section V discusses the evaluation of the *Emo-CSI* framework results with the related works. Finally, Section VI presents the conclusion and the future works of this research.

II. Emotional Detection Principles

This section describes the emotional detection principles. The emotion classification applied in this proposed framework is described in section II-A. To conduct the experiment, the stimuli are applied to stimulate the emotion of the participants. The emotion assessment process is described in section II-B.

A. Emotion Classification

In psychological theories, emotions can be detected by human expressions. Emotions can be recognized by facial, vocal, gesture expressions, and physiological signals [3], [13].

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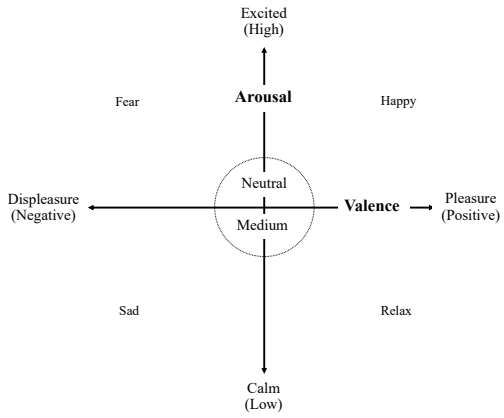

Fig. 1. Valence - Arousal Circumplex Model

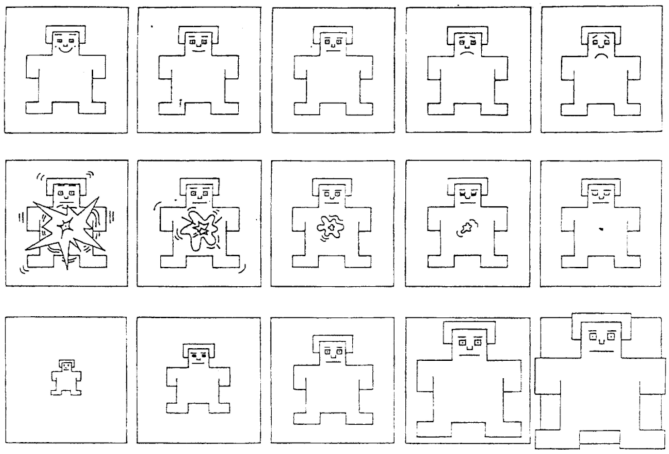

Fig. 2. Pictorial Assessment in SAM

However, facial, vocal and gesture expressions can be controlled to conceal the emotional state [6], [7].

There are several research ideas to describe the emotional state. Paul Ekman [14] proposed six discrete basic emotions which are anger, disgust, fear, happiness, sadness, and surprise. Robert Plutchik [15] developed the wheel emotion model from Ekman's discrete emotions. The model suggests eight primary emotions which are fear, anger, sadness, joy, disgust, surprise, trust, and anticipation. These eight primary emotions aligned on four negative-positive bipolar axises. These bipolar axises are sadness-joy, fear-anger, disgust-trust, and anticipation-surprise.

Moreover, emotions can be classified with multidimensional scaling. Dimensional models of emotion attempt to classify human emotion by defining where the emotions lie on two or three dimensions. The most well-known dimensional model is the circumplex model which was developed by James Russell [16]. This model suggests that emotions are distributed in a two-dimensional circular space, containing arousal and valence dimensions. The center of the circle represents a neutral valence and medium level of arousal. In this model, emotional states can be represented at any level of valence and arousal, or at a neutral level of one or both of these factors. The model represents an emotion by valence (horizontal axis) and arousal (vertical axis) shown in fig. 1. Valence axis represents negative to positive emotion (displeasure-pleasure). Arousal axis represents low to high activation (calm-excited).

This paper applies the concept of James Russell's circumplex model to classify the emotion. The emotion can be scaled on valence and arousal. The emotional classification labels are also suitable for automatic emotion recognition applications.

B. Emotion Assessment Process

To evaluate the emotion classification accuracy, the proposed *Emo-CSI* framework is applied to volunteer participants. The volunteer participants are aroused by the standardized emotional stimuli and self-assess their emotions using standardized emotional assessment.

- *Standardized Emotional Stimuli:* Stimuli (e.g, pictures, sound, music, video clip, and computer game) can elicit emotion [7]. The Center for Emotion and Attention (CSEA) at the University of Florida has developed and distributed a standardized emotional stimuli called the International Affective Picture System (IAPS) [17] and the International Affective Digitized Sound system (IADS)[18]. The IAPS is a set of photographs taken from everyday objects and scenes such as furniture, mutilated bodies, animals, people activities, and erotic nudes. The IADS is a set of sounds from everyday occurrences such as the sounds of animals, environment, people activities, and erotic scenes. These stimuli (IADS and IAPS) have been experimented on participants and become the standardized emotional stimuli. Moreover, it is found that the combination of visual and auditory stimulus make an emotional state stronger than using only visual or auditory stimulus [19], [20]. Thus, this paper applies the combination of IAPS and IADS as the emotional stimuli.
- *Emotional Assessment:* The volunteer participants also provide the self-assessment of their emotion after emotional stimulation. The Self-Assessment Manikin (SAM)[21] is an affective emotion assessment. SAM is a pictorial assessment with non-verbal. SAM is used to directly measure the valence and arousal associated with a person's emotional reaction to the stimulus. This paper applies the SAM to the participant to self-validate their emotions.

III. Design of *Emo-CSI*

The emotional classification framework, called *Emo-CSI* (the ***Emo***tional ***C***lassification using physiological ***S***ensor ***I***nterpretation), has been designed to classify the human emotional state. The *Emo-CSI* framework, as depicted in Fig. 3, consists of three components which are the following: 1) physiological signal sensing 2) data pre-processing, and 3) emotional state classification.

A. Physiological Signal Sensing

The *Emo-CSI* framework applies physiological sensors to detect the physiological signals from the participant who is stimulated by the standardized emotional stimuli. The physiological sensors consist of five measurement units which are controlled by controller unit, as shown in Fig. 3.

1) Heart Rate Measurement Unit: contains the pulse sensor ESEN220 model SEN-11574 [22], manufactured by Sparkfun Electronics. The pulse sensor uses photoplethysmograph (PPG) to measure blood-oxygen level. The heart rate signals

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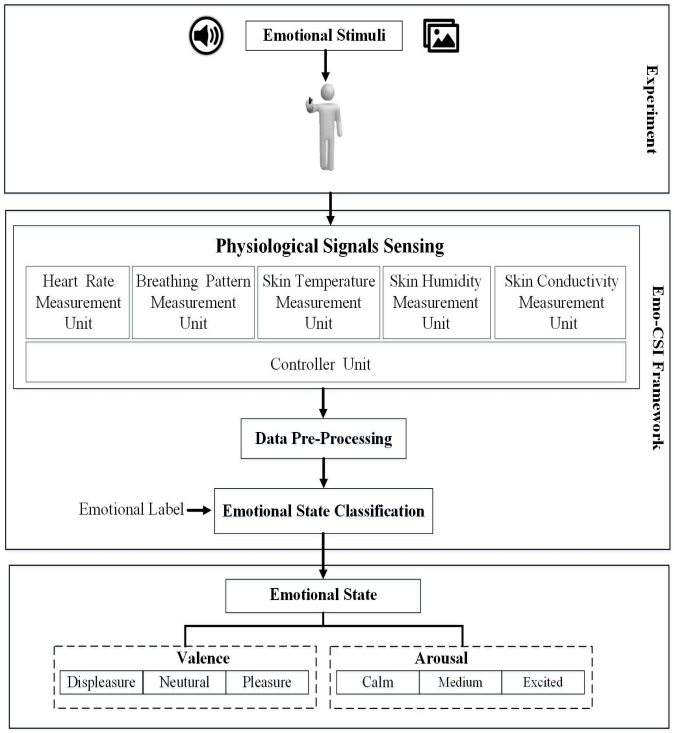

Fig. 3. Emo-CSI Process

are measured by contacting the sensor on the participant's fingertip or earlobe. Fig. 4 shows the heart rate measurement unit used in the experiment.

2) Breathing Pattern Measurement Unit: contains two sensors as shown in Fig. 5. The first sensor uses the toggle switching technique to detect the breathing pattern. The chest belt, included the switching technique sensor, is contacted to the participant. The switch sensor will be pressed when inhaling, whereas when exhaling, the switch sensors will be released. The second sensor uses the valuable resistance technique to measure the voltage. This sensor is attached to the waist belt contacted to the participant. Both techniques are applied for assuring the breathing pattern signals data collection.

3) Skin Temperature Measurement Unit: uses SHT31-A sensor [23], manufactured by ClosedCube, for measuring skin temperature (SKT). The typical accuracy of the sensor are $\pm 2\%$RH and $\pm 0.3°$C. The sensor is attached to left hand side glove and test bed shoes as shown in Fig. 6 and 7.

4) Skin Humidity Measurement Unit: uses SHT31-A sensor [23], as used in skin temperature measurement unit, for measuring skin humidity (SKH).

5) Skin Conductivity Measurement Unit: contains a sensor which uses the valuable resistance technique (as described in subsection III-A2) for measuring skin conductive (SKC). This sensor is attached to left hand side glove and test bed shoes as shown in Fig. 6 and 7

6) Controller Unit: contains an Arduino Mega 2560 microcontroller board based [24] and signal preamplifier circuit. The microcontroller is used to control all measurement sensing units. The preamplifier circuit is used to pre-process the physiological signals.

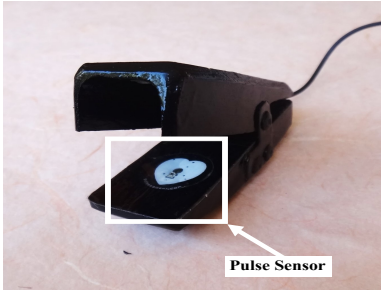

Fig. 4. Heart Rate Measurement Unit

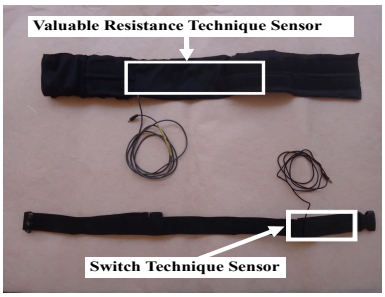

Fig. 5. Breathing Pattern Measurement Unit

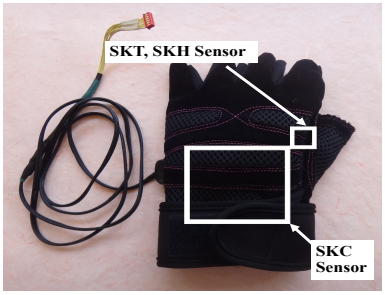

Fig. 6. Hand Set

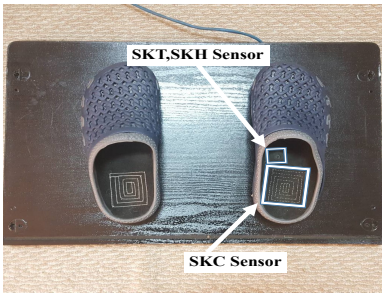

Fig. 7. Foot Set

B. Data Pre-processing

The physiological signals, measured from the physiological signal sensing component, are pre-processed to enter in the emotional state classification component. The physiological signals, measured from the physiological signal sensing component, are pre-processed to enter to the emotional state classification component. The data signals are normalized. Then, the statistical processes are applied to summarize the data signals. The outputs of this component are the average, maximum, minimum and standard deviation of each physiological signal during the period of emotional state session. Thus there are 32 features that will be provided to the emotional state classification component.

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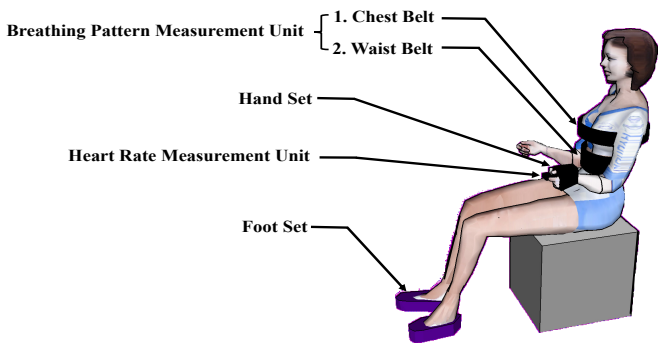

Fig. 8. Sensors Arrangement

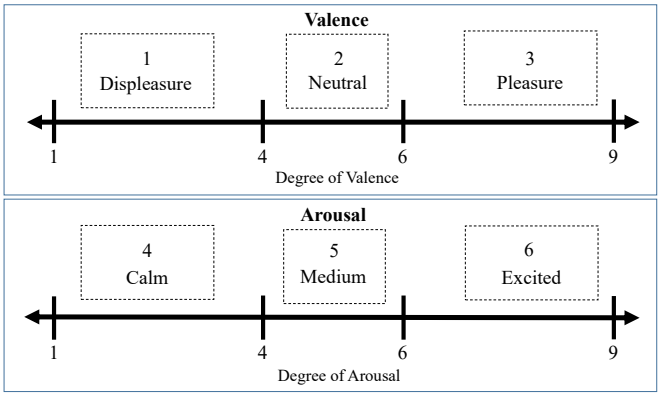

Fig. 9. The Defined Classes in Valence-Arousal Model

C. Emotional State Classification

The emotional state classification component takes the pre-processed signal data and emotional labels for training the classification model. The emotional labels are the standardized emotional classification of emotional stimuli used in the experiment. The stimuli scores on valence and arousal axis are the number between one to nine. These emotional labels are classified to match Valence - Arousal circumplex model as shown in Fig. 1. The model classifies the emotional state to six emotional labels (i.e., Displeasure, Neutral, Pleasure, Calm, Medium, and Excited) by the stimuli score as shown in Fig. 9. In this paper, three emotional classification techniques (i.e., ANN, SVM, and decision tree) are investigated.

1) Artificial Neuron Network (ANN): The artificial neural network is inspired by the human nervous system structure. Artificial neural networks generate nodes (Neuron) and interconnections between nodes (dendrite) to class label (axon) [25].

2) Support Vector Machine (SVM): The support vector machine uses mathematics vectors (hyper-plane) for class splitting. A mathematical model creates the hyper-plane which can be a linear or nonlinear model (e.g., linear, polynomial, sigmoid, radial basis equation) [26].

3) Decision Tree (DT): The decision tree classification model is inspired by the tree branches. The decision tree model starts from a root node and creates branches and leafs. Each branch represents outcome of the test. Each leaf node represents the class label. The paths from root to leafs represent classification rules [27].

The parameters in classification techniques are empirical defined such as the ANN has no intermediate node or hidden layer, the SVM use the linear model, and the DT use classification and regression tree (CART) algorithm for splitting tree branches.

IV. Experiment

This section describes the experiment procedure to evaluate the *Emo-CSI* framework in section IV-A. This section also evaluates the relationship of the physiological sensor signals to the emotional state in section IV-B and investigates the accuracy of the classification techniques in section IV-C.

A. Experiment Procedure

In this experiment, the physiological signals are collected from 23 volunteer participants. The participants are ten females and thirteen males. Participants age between 20-27 years old with the average of 23.87 years old and standard deviation of 2.05. All participants are healthy with no psychiatric disease.

The experiment for each participant takes about half an hour to be completed. The experiment procedure includes two sessions which are 1) the pre-test stimulation session and 2) the stimulation session. The pre-test stimulation session applies to the participant for system checking. Each emotional stimulus is applied to the participant for six seconds. Then, the participant performs the self-assessment (SAM) for 15 seconds. Then, physiological data obtained in the computer are checked. The physiological sensors may be adjusted in this period. Then, the stimulation session is performed. There are three sub-sessions which are the following: 1) baseline state, 2) emotional state, and 3) recovery state. At the baseline state, the blank screen is displayed on the monitor for 15 seconds. In this period, the baseline of physiological signals are measured. At the emotional state, each emotional stimulus includes a picture from IAPS and a matching sound from IADS is played for six seconds. In this period. the physiological signals are measured. Then, the participant performs the self-assessment (SAM) for 15 seconds. This emotional state repeatedly runs for 31 rounds. Finally, at the recovery sate, the blank screen is displayed on the monitor for 15 seconds; the experiment procedure is finished. Fig. 10 depicts this experiment procedure. The physiological signals from emotional state are applied for classification.

B. Physiological Signal Data Evaluation

This section evaluates the physiological signal data in *Emo-CSI* framework. The signal value is obtained by the 10-bit analog converter which ranges between 0-5 volt. The voltage resolution is 4.88 mV. Fig.11 and 12 show the difference of signals between baseline state and stimulus state (stimulus No.26) of participant No.13 in 15 seconds. The figures show that the frequency of HR signal from the baseline state is less than that from the emotional state. The figures show that the breathing patterns (BR1 and BR2) of the baseline state and the emotional state are different. The skin conductivity, temperature, and humidity (SKC, SKT, and SKH) from the emotional state are higher than that from the baseline state. Thus, these figures show that the physiological signals can be detected from physiological sensors. The physical signal data are different when the participant is emotionally stimulated.

C. Classification Result

This section evaluates the accuracy of the classification of *Emo-CSI* in testing data. The accuracy results are divided into

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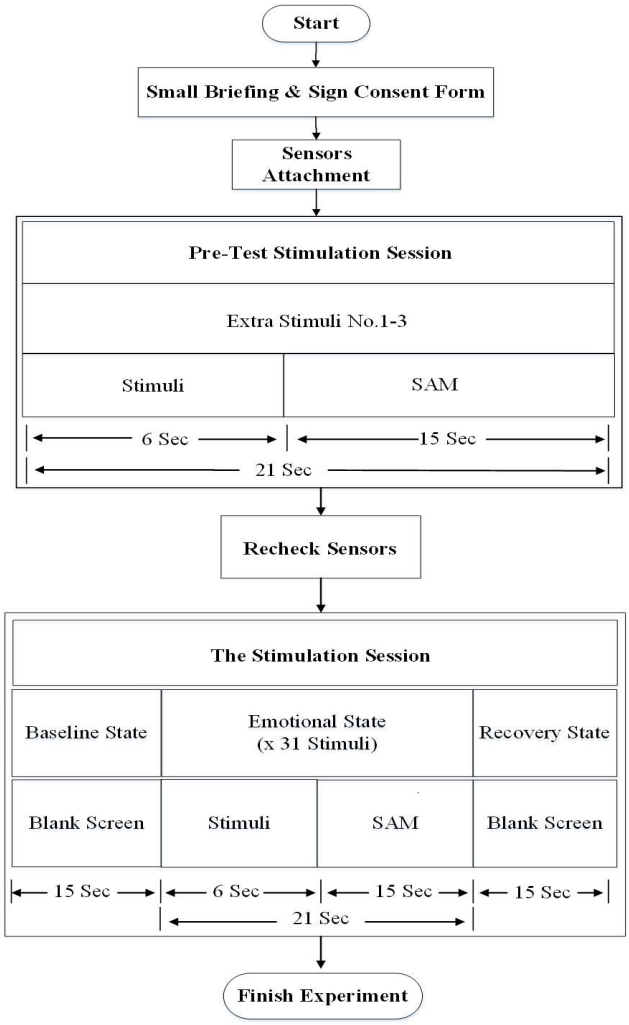

Fig. 10. The Experiment Procedure

valence and arousal axis as shown in Table.I and Table.II. From Table. I, the best accuracy for valence axis is 59.54%. The top three accuracy sensors are SKC from the hand set, HR from the heart set, and SKT from the hand set. From Table. II, the best accuracy for arousal axis is 63.00%. The top three accuracy sensors are SKT from the foot set, SKH from the hand set and HR from the heart set. The best ranking accuracy from valence axis is 57.73% and arousal axis is 60.50%. The combination of all individual sensors accuracy from valence axis is 55.45% and arousal axis is 59.00% . These tables show that all of the best accuracy on valence and arousal axis are obtained by using SVM as the classification.

V. Related Work

There are many emotion recognition researches which use the physiological signals for classify the emotional states in valence-arousal model. These researches use difference stimuli such as IAPS [3], Quran Recitation [28], and audio-visual film clips [11] to elicit the emotional states. The valence-arousal model is used to design emotional states in paper [3], [28] and [7]. In paper [3], the authors defined four emotional states based on valence and arousal model which are high valence high arousal (HVHA), high valence low arousal (HVLA), low valence high arousal (LVHA) and low valence low arousal (LVLA). In paper [28], the authors classify emotional states into four quadrant of valence-arousal model which are fear, happy, sad, and calm. In paper [7], the authors classify six

TABLE I
Comparison Classifier Result for Valence Axis

		Classification Rate of Valence Axis (%)		
		SVM	**ANN**	**DT**
Heart Set		58.64	49.55	45.45
Breathing Set		56.82	49.55	39.09
Hand set	SKT	57.27	51.81	45.91
	SKH	55.45	47.27	48.18
	SKC	**59.54**	51.36	47.27
Foot set	SKT	55.45	51.81	41.36
	SKH	55.90	51.81	45.91
	SKC	57.27	51.81	45.91
The Ranking		57.73	49.09	44.55
All Sensors		55.45	49.54	39.54

TABLE II
Comparison Classifier Result for Arousal Axis

		Classification Rate of Arousal Axis (%)		
		SVM	**ANN**	**DT**
Heart Set		62.00	52.00	49.00
Breathing Set		61.00	53.00	49.00
Hand set	SKT	58.00	54.00	43.00
	SKH	62.50	59.50	52.50
	SKC	60.50	51.50	47.00
Foot set	SKT	**63.00**	51.50	45.00
	SKH	59.50	49.00	42.50
	SKC	59.00	50.50	45.00
The Ranking		60.50	50.00	47.50
All Sensors		59.00	59.00	51.50

emotional states which are displeasure, neutral, pleasure, calm, medium, and excited by using valence and arousal axis. This paper also uses six emotional states which are the same as the paper [7]. The paper [7] uses the MAHNOB-HCI dataset. This dataset collects the response of participant body while they watch the video clips through various sensors, cameras and microphones. The best classification accuracies in the paper [7] are 56.83% for valence and 54.73% for arousal. On the other hand, this paper uses the combination of IAPS and IADS. The best classification accuracies in this paper are 59.54% for valence and 63.00% for arousal.

The physiological signals which are used in emotion recognition researches are measured by the medical or scientific research instrument such as MP100WS of Biopac Systems Inc [11], Zephyr BioHarness, and ADInstruments [3]. This paper shows that the custom-made sensors for emotional states classification, which are low-cost and low-precision compared to the medical or scientific research instruments, are able to measure physiological signals for emotional state classification. The proposed *Emo-CSI* framework is able to yield acceptable accuracy for emotional state classification.

VI. Conclusion

The emotion recognition is important to developing human computer interaction and useful to assist people who cannot express emotion. In this paper, the *Emo-CSI* framework has been presented to apply as an automatic emotion recognition. The *Emo-CSI* framework consists three components which are 1) physiological signal sensing, 2) data pre-processing, and 3) emotional state classification. The experimental results show that the physiological signals from the custom-made sensors can be applied to classify the emotion. The classification result shows that the best accuracy of valence axis is 59.54% and arousal axis is 63.00%. The best accuracy is obtained by using SVM as the classification model.

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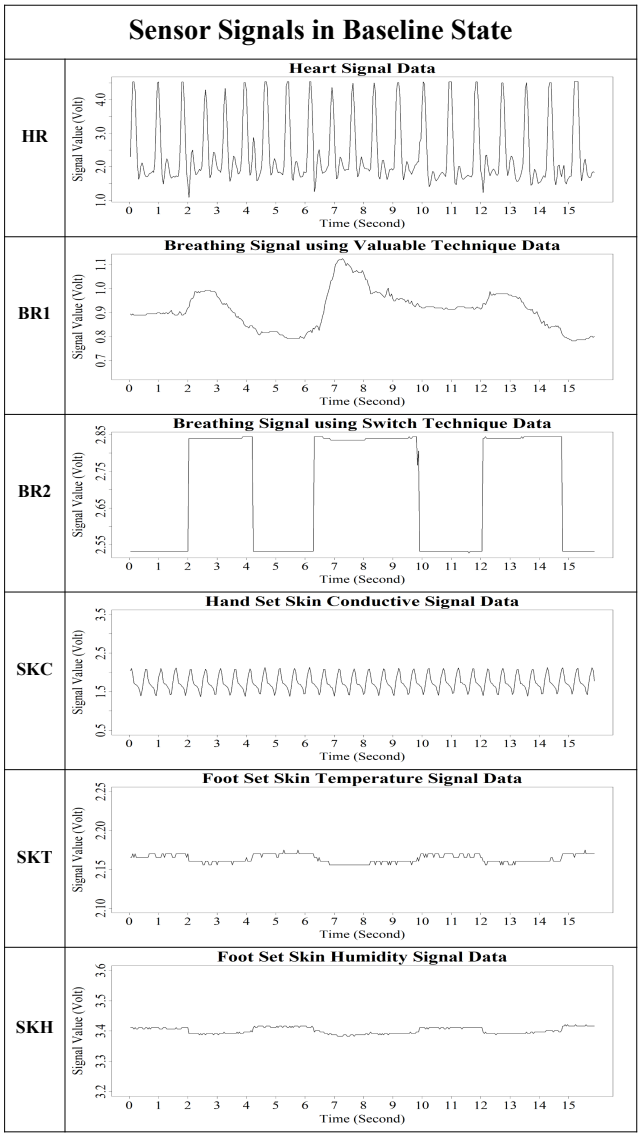

Fig. 11. Baseline Signals

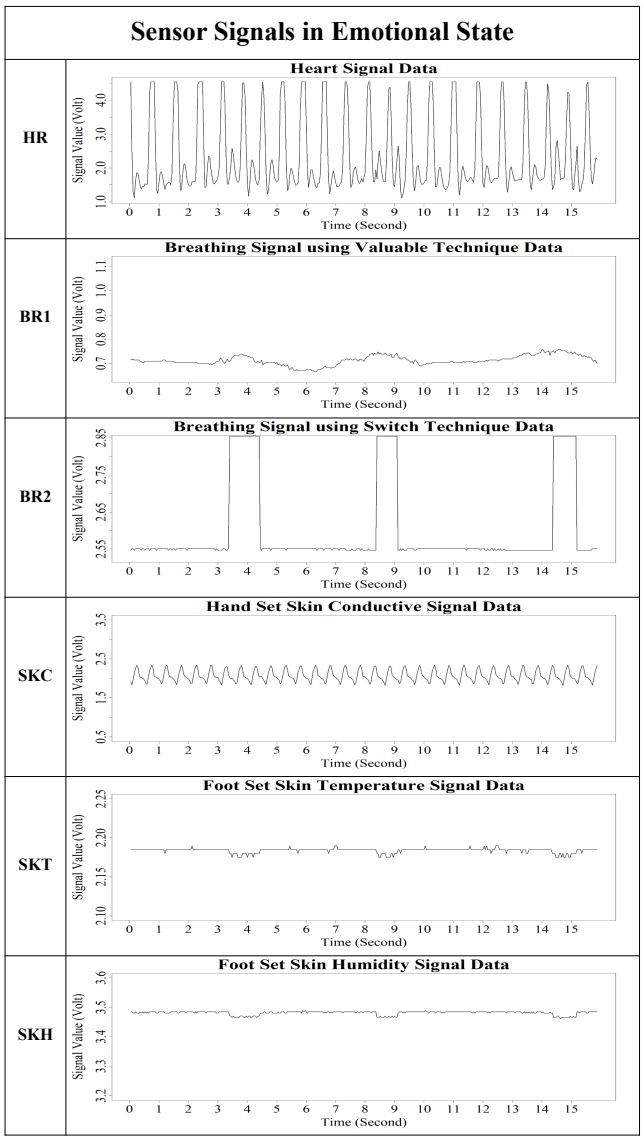

Fig. 12. Stimulus Signals

VII. Future Work

Some future works are planned. The data pre-processing can be improved such as adding a low pass filter to the data signals for noise elimination and applying the fast fourier transform (FFT) to represent the data signals in frequency domain in order to improve classification accuracy. Moreover, the experiment to validate the self-assessment and the standardized emotional assessment can be compared to decreasing social and cultural differences.

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References

[1] W. J. Freeman, "Emotion is essential to all intentional behaviors," in *Emotion, development, and self-organization: Dynamic systems approaches to emotional development*, G. I. Lewis, Marc D, Ed. Cambridge University Press Cambridge, England, 2000, pp. 209–235.

[2] C. Darwin and P. Prodger, *The expression of the emotions in man and animals*. Oxford University Press, USA, 1998.

[3] S. Basu, N. Jana, A. Bag, M. Mahadevappa, J. Mukherjee, S. Kumar, and R. Guha, "Emotion recognition based on physiological signals using valence-arousal model," in *Proc. of Third IEEE International Conference on Image Information Processing (ICIIP)*, 2015, pp. 50–55.

[4] L. Müller, A. Bernin, S. Ghose, W. Gozdzielewski, Q. Wang, C. Grecos, K. von Luck, and F. Vogt, "Physiological data analysis for an emotional provoking exergame," in *Proc. of IEEE Symposium Series on Computational Intelligence (SSCI)*, 2016, pp. 1–8.

[5] T. Hayashi and T. Yoshida, "Impression estimation of a video based on the valence-arousal model," in *Proc. of IEEE International Conference on Multimedia Applications and Signal Processing (ICMSSP)*, 2016, pp. 41–45.

[6] P. Gong, H. T. Ma, and Y. Wang, "Emotion recognition based on the multiple physiological signals," in *Proc. of IEEE International Conference on Real-time Computing and Robotics (RCAR)*, 2016, pp. 140–143.

[7] M. B. H. Wiem and Z. Lachiri, "Emotion sensing from physiological signals using three defined areas in arousal-valence model," in *Proc. of IEEE International Conference on Control, Automation and Diagnosis (ICCAD)*, 2017, pp. 219–223.

[8] J. A. Soto and R. W. Levenson, "Emotion recognition across cultures: The influence of ethnicity on empathic accuracy and physiological linkage," *Journal of Emotion*, vol. 9, no. 6, p. 874, 2009.

[9] P. Ekman, "Are there basic emotions?" *Journal of Psychological Review*, vol. 99, no. 3, p. 550, 1992.

[10] D. Matsumoto, "Methodological requirements to test a possible in-group advantage in judging emotions across cultures: Comment on elfenbein and ambady and evidence," *Journal of Psychological Bulletin*, vol. 128, no. 2, p. 236, 2002.

[11] E.-H. Jang, B.-J. Park, S.-H. Kim, Y. Eum, and J.-H. Sohn, "A study

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on analysis of bio-signals for basic emotions classification: recognition using machine learning algorithms," in *Proc. of IEEE International Conference on Information Science and Applications (ICISA)*, 2014, pp. 1–4.

[12] K. Gouizi, F. Bereksi Reguig, and C. Maaoui, "Emotion recognition from physiological signals," *Journal of medical engineering & technology*, vol. 35, no. 6-7, pp. 300–307, 2011.

[13] E.-H. Jang, B.-J. Park, S.-H. Kim, Y. Eum, and J.-H. Sohn, "Identification of the optimal emotion recognition algorithm using physiological signals," in *Proc. of IEEE International Conference on Engineering and Industries (ICEI)*, 2011, pp. 1–6.

[14] P. Ekman, "An argument for basic emotions," *Journal of Cognition & emotion*, vol. 6, no. 3-4, pp. 169–200, 1992.

[15] R. Plutchik, "The nature of emotions human emotions have deep evolutionary roots, a fact that may explain their complexity and provide tools for clinical practice," *Journal of American scientist*, vol. 89, no. 4, pp. 344–350, 2001.

[16] J. Posner, J. A. Russell, and B. S. Peterson, "The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and psychopathology," *Journal of Development and psychopathology*, vol. 17, no. 3, pp. 715–734, 2005.

[17] P. J. Lang, M. M. Bradley, and B. N. Cuthbert, "International affective picture system (iaps): Affective ratings of pictures and instruction manual. technical report a-8," University of Florida, Gainesville, FL, Tech. Rep., 2008.

[18] P. Lang, M. Bradley, and B. Culthbert, "The international affective digitized sounds (2nd edition; iads-2): Affective ratings of sounds and instruction manual," University of Florida, Gainesville, Fl, Tech. Rep., 2007.

[19] A.-M. Brouwer, N. Van Wouwe, C. Mühl, J. van Erp, and A. Toet, "Perceiving blocks of emotional pictures and sounds: effects on physiological variables," *Journal of Frontiers in human neuroscience*, vol. 7, 2013.

[20] K. N. Spreckelmeyer, M. Kutas, T. P. Urbach, E. Altenmüller, and T. F. Münte, "Combined perception of emotion in pictures and musical sounds," *Journal of Brain research*, vol. 1070, no. 1, pp. 160–170, 2006.

[21] M. M. Bradley and P. J. Lang, "Measuring emotion: the self-assessment manikin and the semantic differential," *Journal of behavior therapy and experimental psychiatry*, vol. 25, no. 1, pp. 49–59, 1994.

[22] S. Electronics. (2015) Pulse Sensor Getting Started Guide. [Online]. Available: https://www.allaboutcircuits.com/electronic-components/datasheet/SEN-11574-Spark-Fun

[23] S. Company. (2015) Datasheet SHT3x-ARP. [Online]. Available: https://www.mouser.com/ds/2/682/Sensirion_Humidity_Sensors _SHT3x _Datasheet_analog-1079681.pdf

[24] D. Mellis. (2011) Arduino Mega 2560 Datasheet. [Online]. Available: http://www.mantech.co.za/datasheets/products/A000047.pdf

[25] R. J. Schalkoff, *Artificial neural networks*. McGraw-Hill New York, 1997, vol. 1.

[26] C. Cortes and V. Vapnik, "Support vector machine," *Journal of Machine learning*, vol. 20, no. 3, pp. 273–297, 1995.

[27] J. R. Quinlan, "Induction of decision trees," *Journal of Machine learning*, vol. 1, no. 1, pp. 81–106, 1986.

[28] S. A. Y. Al-Galal, I. F. T. Alshaikhli, A. W. bin Abdul Rahman, and M. A. Dzulkifli, "Eeg-based emotion recognition while listening to quran recitation compared with relaxing music using valence-arousal model," in *Proc. of 4th IEEE International Conference on Advanced Computer Science Applications and Technologies (ACSAT)*, 2015, pp. 245–250.

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